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Bridging Conflicts and Biodiversity Protection

The Critical Role of Reliable and Comparable Data

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Abstract

Biodiversity is essential for ecological stability, human well-being, and economic progress, providing critical ecosystem services such as clean water, food, and climate regulation. However, it faces unprecedented threats, with extinction rates accelerating to 1,000 times the natural baseline due to habitat destruction, overexploitation, pollution, invasive species, illegal trade, and climate change. Effective conservation requires urgent, coordinated global action, as ecosystems and species habitats often transcend national borders. Collaboration among governments, industries, and communities is essential to restore habitats, protect endangered species, strengthen policies, and enforce conservation measures. The challenges of biodiversity conservation are particularly acute in geopolitically sensitive and overlapping regions, including non-determined legal status territories, fragile and conflict-affected situations, and transboundary ecosystems. In these areas, effective conservation is hindered by weak policies, inconsistent enforcement, and institutional fragility. This paper addresses these challenges by providing baseline data to guide conservation strategies. Using newly developed World Bank species occurrence maps based on open-access, date-stamped records from the Global Biodiversity Information Facility, the study evaluates species richness, endemism, and extinction risks across 35 non-determined legal status territories, 19 conflict-affected countries, 20 fragile states, 18 marine joint regimes, and 311 international river basins. The data sets reveal that these regions host numerous, often vulnerable, species. Biodiversity conservation emerges as a pathway for trust-building and collaboration, aligning stakeholders around shared goals such as climate resilience and sustainable livelihoods. Reliable and comparable data sets are critical for evidence-based planning, fostering dialogue and cooperation among divided groups. The estimates presented in this paper aim to support robust, data-driven strategies to safeguard biodiversity in geopolitically sensitive and overlapping regions, with far-reaching implications for global conservation and international cooperation.

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Bridging Conflicts and Biodiversity Protection: The Critical Role of Reliable and Comparable Data

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Introduction

Biodiversity conservation is essential for sustainable development, poverty reduction, and a healthy planet. It supports ecosystems that are fundamental to human livelihoods by providing food, clean water, and climate stability. Healthy ecosystems ensure the availability of natural resources critical to agriculture, fishing, and forestry, helping to alleviate poverty and promote sustainable economic growth. Additionally, biodiversity preservation supports industries like ecotourism and pharmaceuticals, which generate revenue and create jobs, further contributing to shared prosperity. Maintaining biodiversity is also a key to a livable planet, as it strengthens the resilience of ecosystems that regulate the climate by sequestering carbon and mitigating the impacts of climate change. Ecosystems such as forests, wetlands, and oceans act as natural buffers, stabilizing global temperatures and sustaining life on Earth. Thus, investing in biodiversity conservation is crucial not only for environmental health, but also for human well-being and global sustainability.

However, biodiversity is declining at an alarming rate. A recent UN report by the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services revealed that 1 million species are near extinction, with the rate of extinctions accelerating despite increasing awareness of the interconnectedness between human life and nature (IPBES 2019). This report, based on a systematic review of approximately 15,000 scientific and governmental sources, aligns with the findings of the Living Planet Index,[1] which has recorded a 68% decline in global biodiversity since 1970. The primary drivers of this crisis include habitat destruction, overexploitation of resources, pollution, illegal wildlife trade, invasive species and climate change.

In response to these unprecedented losses, immediate and coordinated international action is required. Every country and territory must actively engage in biodiversity conservation to halt further biodiversity decline. Nations need to implement effective policies and conservation strategies tailored to local ecosystems, while leveraging global frameworks such as the Convention on Biological Diversity (CBD) and the United Nations Sustainable Development Goals (SDGs).

One of the significant barriers to effective conservation is the impact of human conflicts, particularly in weakly-governed areas. These regions often experience destructive effects on biodiversity, spanning terrestrial, marine, and freshwater ecosystems. Physical damage to landscapes, habitat destruction, pollution, and overexploitation of natural resources are common consequences of such conflicts. Military dimensions of conflicts can further exacerbate these challenges by diverting resources from environmental protection and disrupting natural food webs and migratory patterns (Rist, Norstrom, and Queiroz 2024; Parkinson and Cottrell 2022). The academic literature extensively documents the challenges of biodiversity conservation within

[1] https://www.livingplanetindex.org/

non-determined legal status territories and conflict zones (e.g., Greiner 2012; Daskin and Pringle 2018; Hanson 2018).

Moreover, weak governance and fragmented authority in non-determined legal status territories create additional complications. Conflicting territorial claims can undermine environmental protection and allow unchecked exploitation of natural resources. The institutional landscape in these areas often features competing authorities, which results in inconsistencies in law enforcement and resource management (Beckman, 2013). Although international interventions aim to foster stability, they frequently lack robust enforcement mechanisms. This instability can prolong conflicts, deepen socio-economic inequalities, and intensify environmental degradation (Buhaug et al., 2014).

In conflict zones, governance often collapses, making effective environmental protection nearly impossible. For instance, recent events in the Syrian Arab Republic and the Republic of Yemen have led to the collapse of public institutions, with attention shifting to immediate survival and security needs, leaving environmental concerns neglected (World Bank, 2011). To address these challenges, efforts to conserve biodiversity in affected regions must be preceded by the restoration of national and international cooperation to rebuild trust and establish shared conservation goals (Gaind et al., 2016).

While human conflict areas present significant challenges for biodiversity conservation due to governance gaps and instability, transboundary river basins also face complex issues—ranging from competing national interests to opportunities for cooperation—requiring coordinated efforts to address shared environmental concerns and manage resources sustainably across borders. Effective conservation in such areas requires objective, reliable, and comparable data from all riparian entities. In practice, weak governance and political or social instability often make it difficult for officially-designated institutions to collect the necessary data (Cohen and Arieli 2011). This is particularly true in regions with competing territorial claims, where data collection and conservation efforts are frequently obstructed. In this context, recent studies (Turner et al., 2015; Levin et al., 2019; Mobaied and Rudant 2019; Garzón and Valánszki 2020; Aung 2021) have emphasized the potential importance of remote sensing from satellite platforms. These technologies offer a promising solution by providing valuable and consistently collected real-time data from areas that may otherwise be inaccessible or where traditional monitoring is too challenging.

The Global Biodiversity Information Facility (GBIF) is also playing a critical role in overcoming these barriers. By acquiring and distributing data on species sightings from concerned organizations and individuals, the GBIF is providing essential information in areas where officially-supported monitoring may be sparse or difficult to implement. This collaboration strengthens global conservation efforts by making vital data more accessible, even in fragile or conflict-affected regions.

To implement effective conservation strategies in such contexts, accurate, location-specific data are crucial. The importance of this data for identifying critical biodiversity areas and tracking conservation progress has been highlighted in numerous studies. However, inconsistencies in data across countries and administrative boundaries hinder the accurate assessment of ecosystem health and biodiversity trends (Urbano et al., 2024; Geijzendorffer et al., 2016; Pereira et al., 2013). Moreover, the growing gap between emerging threats and infrequently updated data, often managed by underfunded institutions, presents an ongoing challenge.

To help address this gap, we have produced new public datasets by utilizing millions of georeferenced species occurrence records from the Global Biodiversity Information Facility (GBIF) (Dasgupta et al. 2024a). Using machine-based pattern recognition, we developed a comprehensive database covering the occurrence regions of nearly 600,000 species. These maps were validated by comparing them with expert-reported maps for mammals, ants, and vascular plants, revealing strong correlations in global distribution patterns. Our expanded database includes a broad array of species across terrestrial, freshwater, and marine environments, encompassing arthropods, mollusks, invertebrates, plants, fungi, and more, alongside traditionally studied amphibians, birds, fish, reptiles, and mammals. Figure 1 illustrates the composition of this expanded database, revealing new global distribution patterns that can inform conservation planning.

In this paper, we focus on the implications of our results for biodiversity protection in transboundary regions,[2] joint marine regions,[3] non-determined legal status areas,[4] and fragile and conflict-affected situations areas (FCS).[5] Conflicts and coordination failures in such areas have numerous direct and indirect impacts on biodiversity (see Rist, Norstrom and Queiroz (2024) for a review of the relationships between biodiversity, peace and conflict). These are often difficult to address in non-determined legal status areas and FCSs because of weak governance, limited administrative capacity, and insufficient institutions and policies. The effective protection of biodiversity in transboundary areas must rely on cooperation among the parties involved.

[2] A transboundary area is an area of land and/or sea that straddles one or more boundaries between states, sub-national units such as provinces and regions, autonomous areas, and/or areas beyond the limits of national sovereignty or jurisdiction.

[3] A joint marine regime governs shared Exclusive Economic Zones (EEZs) https://www.marineregions.org/eezlinetype.php#:~:text=Joint%20regime,exploitation%20of%20natural%20marine%20resources.

[4] A non-determined legal status area is an area claimed by two or more countries without a determined legal status. Disputed areas can arise from a variety of factors, including the historical background, different religious or ethnic perspectives, possession of strategic natural resources, and fundamental changes in domestic and international environments.

[5] The World Bank has recently released a list of Fragile and Conflict-Affected Situations. The conflict-affected countries are Afghanistan, Burkina Faso, Cameroon, the Central African Republic, the Democratic Republic of Congo, Ethiopia, Haiti, Iraq, Lebanon, Mali, Mozambique, Myanmar, Niger, Nigeria, Somalia, South Sudan, Sudan, Syria, Ukraine, the West Bank and Gaza (territory), and Yemen. Additionally, countries facing institutional and social fragility include Burundi, Chad, the Comoros, the Republic of Congo, Eritrea, Guinea-Bissau, Kiribati, Kosovo, Libya, the Marshall Islands, the Federated States of Micronesia, Papua New Guinea, São Tomé and Príncipe, the Solomon Islands, Timor-Leste, Tuvalu, the República Bolivariana de Venezuela, and Zimbabwe.

Preserving habitats for endemic species, those with restricted ranges, and species at high extinction risk requires special conservation measures tailored to vulnerable and geopolitically complex regions.

The expansion of species information using GBIF records offers a significant expansion of the objective, comparable and reliable data that are critical for successful implementation of transboundary and fragile/conflict state conservation measures. In this paper, we illustrate the potential contribution of our GBIF database data with results for 38 non-determined legal status area areas, 19 conflict-affected countries, 19 countries with pronounced institutional and social fragility, and 311 international river basins.

Figure 1: Total count of species

Species [Class/Order]	Group	Count
Vertebrates [Class] 50,825	Amphibians	5,026
	Birds	10,845
	Mammals	4,377
	Reptiles	7,544
	Fish	23,033
Arthropods [Order] 209,908	Araneae	10,307
	Coleoptera	43,660
	Diptera	23,321
	Hemiptera	13,123
	Hymenoptera	25,924
	Lepidoptera	50,037
	Other	43,536
Vascular Plants [Order] 229,595	Asparagales	17091
	Asterales	22841
	Caryophyllales	9321
	Ericales	8545
	Fabales	16194
	Gentianales	13317
	Lamiales	16436
	Malpighiales	12632
	Myrtales	10253
	Poales	16,433
	Other	86,532
Fungi		**37,450**
Non-animal, non-plant species: protozoans, bacteria, etc.		**16,540**
Other Animals: Molluscs, etc.		**53,214**
Total		**597,532**

Data and Method

The data for this study have been provided by the Global Biodiversity Information Facility (GBIF), a global network, funded by various national governments, that offers open access to comprehensive data on global biodiversity. The GBIF provides a repository of georeferenced, time-stamped species occurrence records that are constantly updated in collaboration with the Catalogue of Life partnership (https://www.catalogueoflife.org), Biodiversity Information Standards (https://www.tdwg.org/), Consortium for the Barcode of Life (CBOL) (http://www.barcodeoflife.org/), Encyclopedia of Life (EOL) (http://eol.org), Global Earth Observation System of Systems (GEOSS) (https://earthobservations.org/geoss.php) and other organizations. The GBIF records span multiple taxa, including animals, plants, fungi, and microbes. Users can directly access these records through GBIF's Occurrence and Maps APIs (https://www.gbif.org/developer/occurrence, https://www.gbif.org/developer/maps), and access to the database is also provided by cloud platforms like Google BigQuery and Amazon Web Services (AWS).

For the analysis presented here, we processed large-scale datasets using Google BigQuery, which efficiently handles vast amounts of biodiversity data. The GBIF occurrence records were filtered to include only those that met specific criteria: we selected records from 1970 onwards and focused on species that had at least three unique reporting locations and five occurrence reports. Additionally, to manage computational efficiency, we limited species with excessive data, such as the American Robin (*Turdus migratorius*), to a maximum of 20,000 randomly sampled records. This filtering approach ensured that computational resources were effectively used, without sacrificing the integrity of the data.

We adhered to GBIF's occurrence reporting protocols, which ensure consistency in the way species records are documented and shared across the global community. This adherence to protocol is essential for maintaining the reliability and comparability of biodiversity data.

Species Occurrence Mapping

To generate species occurrence maps, we employed machine-based techniques for pattern recognition and cluster analysis. For most species (93.6% of the dataset), we applied a spatial boundary construction method known as the alphahull algorithm (Pateiro-López and Rodríguez-Casal, 2010). This method, well-established in the literature (Guo et al., 2022; Kass et al., 2022), allowed for the precise delineation of species' occurrence regions across terrestrial, freshwater, coastal, and marine environments. For the remaining species, we utilized a k-means clustering algorithm combined with convex hulls to estimate their geographic distributions.

To validate the accuracy of our generated maps, we compared them with expert-verified maps from well-regarded datasets of mammals (Marsh et al., 2022), ants (Kass et al., 2022), and vascular plants (Borgelt et al., 2022). These comparisons demonstrated strong correlations between our maps and those generated by subject-matter experts. Figure 2 provides two examples of species occurrence regions: one for an endemic species, *Scherya bahiensis*, which is confined to a small region in Brazil, and one for a non-endemic species, *Lagidium viscacia* (the Mountain Viscacha), which has a broader distribution across several countries in South America. These examples illustrate the contrasting distribution patterns of species with restricted versus extensive geographic ranges.

To illustrate, effective biodiversity conservation plans depend on accurately identifying regions with species that are either endemic (i.e., species confined to a specific geographic area, often within a single country) or facing high extinction risks. Extinction risk refers to the likelihood that a species will become extinct within a certain time frame and is quantified using various models. In this study, we estimate extinction risk using an ordered-logit model, a statistical technique that predicts categorical outcomes based on multiple variables. For this analysis, a high extinction risk is defined as a score above 80 out of 100, indicating a species with a significant threat of extinction.

Utilizing our newly developed occurrence maps, we identified 267,522 endemic species (44.7% of the total species in our dataset) and 84,627 species whose small occurrence areas (defined as areas smaller than 25 km^2) make them highly vulnerable to extinction. Occurrence area refers to the observed geographic range of a species, with smaller ranges often correlating with higher extinction risk due to habitat loss, climate change, or other threats.

In addition, hundreds of thousands of species in our Global Biodiversity Information Facility (GBIF) database have not previously been assessed for extinction risks. To address this gap, we have incorporated location-specific threats (e.g., habitat destruction, pollution, or invasive species) and protection measures (e.g., conservation efforts or legal protections) into the extinction risk estimates for nearly 600,000 species. This includes 510,090 species that have not yet been evaluated for risk by the International Union for Conservation of Nature (IUCN) (Dasgupta et al. 2024b).[6]

Geographic Boundaries for Conservation Analysis

In addition to generating species occurrence maps, we overlaid various geographic boundaries to examine the influence of governance and territorial issues on biodiversity conservation. These boundaries included areas such as non-determined legal status areas, Fragile and Conflict-Affected Situations (FCSs), joint regimes of Exclusive Economic Zones (EEZs), and transboundary river basins, all of which present unique challenges for effective conservation. The boundaries for non-determined legal status areas and fragile/conflict states (FCSs) were obtained from the World Bank. The boundaries of joint EEZ regimes were sourced from the Flanders Marine Institute (2019), and the transboundary river basin boundaries were taken from McCracken and Wolf (2019), version 2022-03-07.

The inclusion of non-determined legal status territories is critical because these regions often face unclear or disputed governance, which can hinder effective conservation efforts. These

[6] Global Biodiversity Species Extinction Risks Data

areas may lack the institutional frameworks necessary for coordinated conservation activities, leading to unchecked exploitation of natural resources and increased threats to biodiversity. In some cases, these regions may be rich in biodiversity, but their legal ambiguity complicates the establishment of conservation policies and enforcement mechanisms.

Fragile and Conflict-Affected Situations (FCSs) are also included due to the significant impact that political instability, civil unrest, and war can have on environmental protection. Biodiversity conservation in FCSs is particularly challenging, as the breakdown of governance structures often leads to deforestation, poaching, illegal mining, and other forms of environmental degradation. Moreover, these regions frequently lack the resources to prioritize environmental protection amid immediate humanitarian and security concerns, further exacerbating threats to biodiversity.

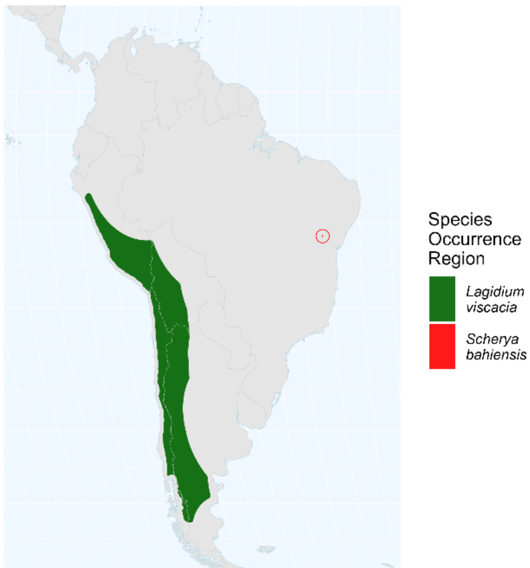
Additional elements for our analysis are provided by terrestrial and marine water bodies that cross national boundaries. Their ecosystems are critical for biodiversity, but are often excluded from shared arrangements for water resource management. Even in cases where ecosystems are included, their protection depends upon monitoring and enforcement provisions that require sustained support.

Our terrestrial analysis focuses on transboundary river basins, which are particularly vulnerable to unsustainable water management practices, pollution, and habitat loss. Managing transboundary river basins requires international cooperation to ensure the long-term health of their ecosystems. Conflicting national interests, such as competing demands for water, can complicate conservation efforts, making these regions particularly important to study.

Our marine analysis focuses on transboundary overlaps in Exclusive Economic Zones (EEZs), which are formally managed by marine joint regimes with specific arrangements for the exploration and exploitation of natural marine resources (Flanders Marine Institute, 2019 and sourced from the United Nations' DOALOS repository). Like transboundary river basin arrangements, these regimes can be inconsistent in their attention to biodiversity management, particularly for migratory marine species. Cooperative governance in these joint regime areas is critical for ensuring that marine resources are used sustainably while safeguarding the biodiversity that exists in shared marine ecosystems.

By overlaying these critical geopolitical boundaries onto our species occurrence maps, we aim to identify areas where conservation efforts may be more difficult due to political, legal, or social challenges. These areas are often at the intersection of biodiversity hotspots and governance weaknesses, making them crucial for targeted conservation actions and policy interventions.

Figure 2: Selected species occurrence regions



Results

Table 1: Area, human population count (WorldPop) and total species in geopolitically sensitive and overlapping areas

	Area (million square Kilometers)	Human Population (millions)	Total species[7] (thousands)
Non-determined legal status Territories	4.6	2.1	55.8
Fragile and Conflict-Affected Situations	37.2	936	123.6
Marine Joint Regimes	0.4	0	61.8
Transboundary River Basins	62	3,500	369.2

[7] Total species is the number of unique species in each geographic category. Some species occur in more than one category.

Taken together, non-determined legal status territories, areas with fragile and conflict-affected situations status, joint marine regimes, and transboundary river basins cover significant portions of the world's terrestrial and marine areas. Non-determined legal status territories cover approximately 4.6 million square kilometers, while areas with fragile and conflict-affected status extend over 37 million square kilometers. Figure 3 shows the geographic distribution of FCS countries across the world. International joint marine regimes cover over 400,000 square kilometers, and transboundary river basins encompass a vast area of nearly 62 million square kilometers. These regions, often characterized by geopolitical tensions and environmental challenges, have substantial importance for international cooperation and conflict resolution. According to population data from WorldPop (Tatem 2017; WorldPop and CIESIN 2018), an estimated 2.1 million people reside within non-determined legal status territories, while around 936 million people live in areas marked by fragile and conflict situations. Furthermore, over 3.5 billion people inhabit regions dependent on shared water resources in transboundary river basins, underlining the complex interplay between population distribution and resource management in these geopolitically sensitive and overlapping areas. Figure 4 illustrates the log human population density in transboundary river basins across the world.

The Role of Human Population Density

Understanding the relationship between biodiversity and human population density is crucial for addressing the ongoing biodiversity crisis. As human populations grow and expand, the pressure on ecosystems intensifies, making it essential to explore how population density impacts biodiversity conservation. Human activities, particularly in densely populated areas, have been a major driver of ecosystem alteration, from habitat loss and fragmentation to overexploitation of resources. The clearing of forests and wetlands for agriculture, urban development, and infrastructure projects has directly contributed to the decline in species diversity and the disruption of critical ecological processes.

Human population density often correlates with higher levels of resource consumption, which in turn often leads to greater environmental degradation. For example, densely populated urban areas tend to have higher pollution levels, increased waste production, and more extensive land conversion. These factors not only stress the natural environment, but also create challenges for species survival. As human settlements encroach on natural habitats, conflicts between humans and wildlife frequently emerge, exacerbating threats to biodiversity. Farmers may face crop damage from wildlife, leading to retaliatory killings, while wildlife may lose their habitats or be forced to migrate into more dangerous or fragmented areas. These interactions highlight the urgent need to assess how human population density contributes to habitat loss and species endangerment.

At the same time, understanding population density's effects on biodiversity can inform strategies for fostering human-wildlife coexistence. Regions with higher population densities require innovative solutions, such as effective land-use planning, the establishment of wildlife corridors, and community-based conservation initiatives. These approaches can help mitigate the impact of human activities on ecosystems, support sustainable agricultural practices, and reduce conflict. Engaging local communities in conservation efforts—particularly in high-density areas—can enhance stewardship, promote sustainable land-use practices, and encourage the protection of critical habitats.

Ultimately, understanding the dynamics between human population density and biodiversity is essential for developing targeted strategies that address the complex interactions between people and nature. By evaluating the impacts of human settlements on ecosystems and promoting sustainable coexistence, we can create a more balanced approach to development and conservation that benefits both biodiversity and human well-being.

Figure 3: Countries determined as Fragile and Conflict-affected Situations areas (FCS) in 2024

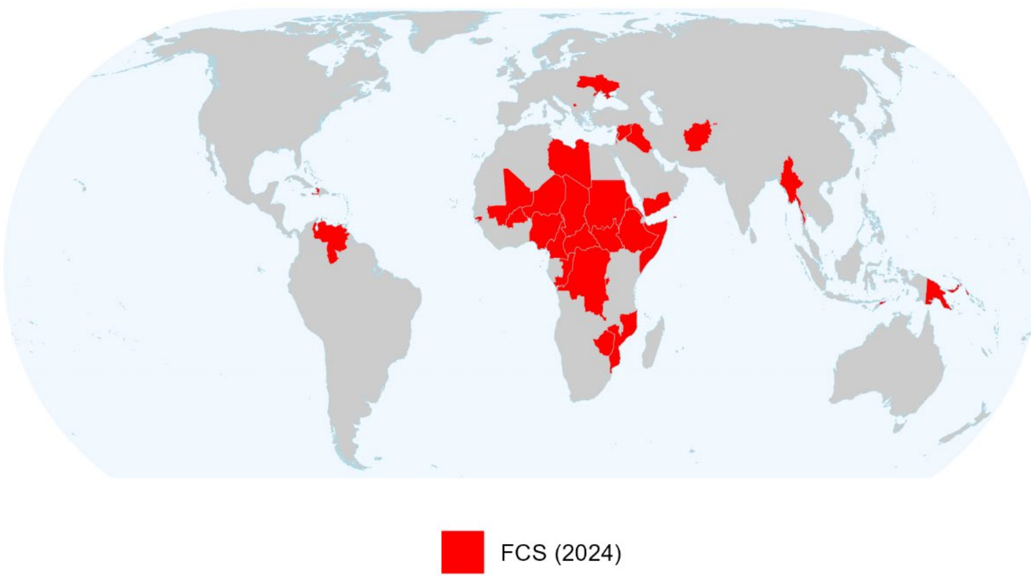


Figure 4: Log human population density in transboundary river basins

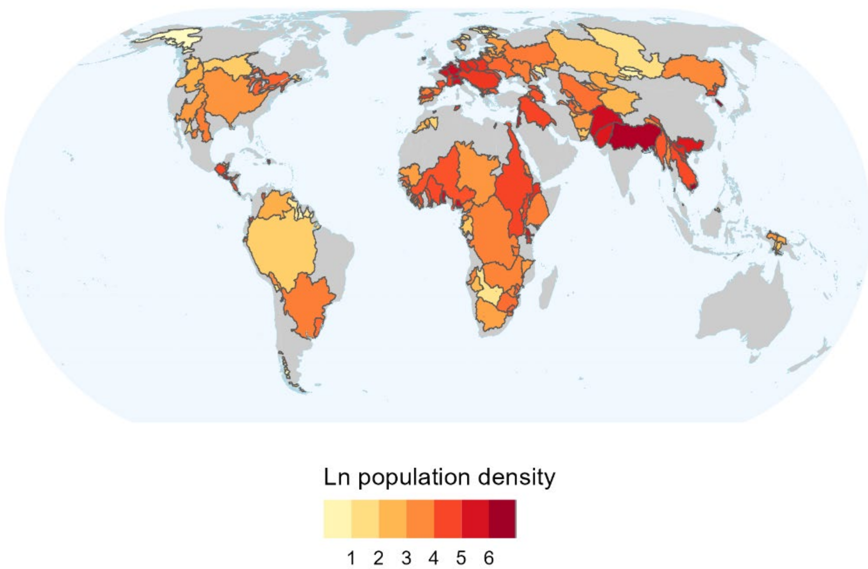
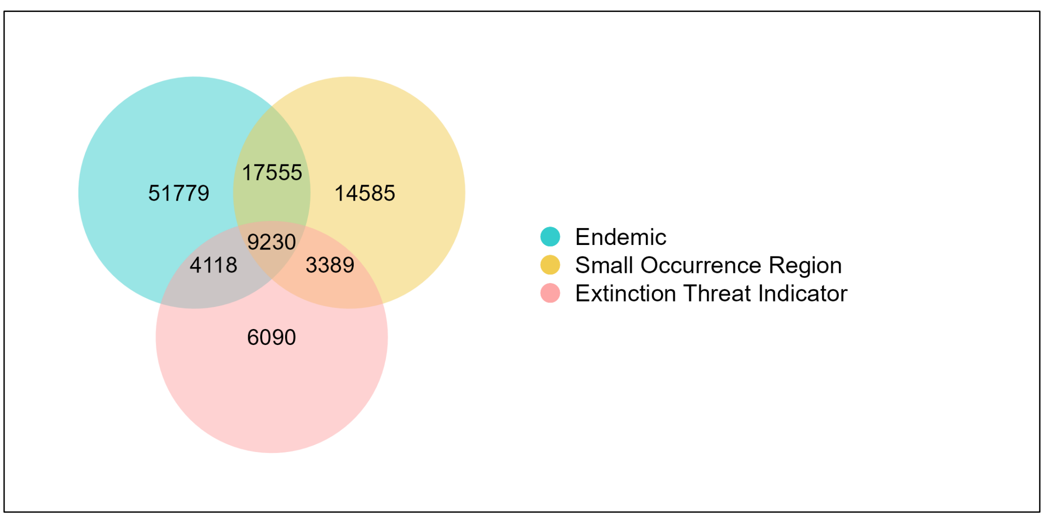


Figure 5: Total count of critical species in geopolitically sensitive and overlapping areas



As an aid to interpreting our results, the Venn diagram in Figure 5 provides an accounting for species in our database that are particularly critical for biodiversity analysis because they are endemic, have small habitats, or have extinction threat indicator values that exceed 80%. The figure shows that while many species are uniquely assigned to one of the three categories, others are in two or three. Among the nearly 600,000 species in our GBIF database, 51,779 have unique endemic assignment, 14,585 have unique small occurrence region assignment, and 6,090 have unique high extinction threat assignment. Among species with non-unique assignments, 17,555 are endemic with small occurrence regions; 3,389 have small occurrence regions and extinction threat indicators higher than 80%; 4,118 are endemics with extinction threat indicators greater than 80%, and 9,230 are in all three categories.[8]

As Table 2 and Figure 5 show, species in all three categories are amply represented in transboundary areas and Fragile and Conflict-Affected Situations. Overall, Table 2 shows that the number of species with potential concern is greatest in transboundary river basins (369,211), followed by fragile/conflict states (123,646), marine joint regimes (58,792) and non-determined legal status territories (55,641). Table 2 also shows that representation is very different by category. Species with small occurrence regions present the biggest challenge in non-determined legal status territories, fragile/conflict states and marine joint regimes, while endemic species dominate in transboundary river basins.

Table 2: Species richness and count of critical species by geopolitically sensitive and overlapping areas

	Count of Total Species (Presence)	Count of Endemic Species	Count of Species with Small Occurrence Region	Count of Species with Threats of Extinction more than 80%
Non-determined legal status Territories	55,641	123	3,238	854
Fragile and Conflict - Affected Situations	123,646	8,649	10,793	3,958
Marine Joint Regimes	58,792	0	2,684	733
Transboundary River Basins	369,211	78,800	40,951	21,362

Figure 6 provides a summary perspective on the information in Figure 5 and Table 2. In the figure, the interior circles are labeled with the estimated total numbers of species in geopolitically sensitive and overlapping areas that are endemic, have small habitats, or have extinction probability indicators above 80%. The oblong extensions for each category are labeled with estimated total species in these categories that are not in geopolitically sensitive and overlapping

[8] We should note that the data in Figure 1 are global in scope. Our discussion of these species in the context of transboundary areas and fragile/conflict states should not be taken to imply that they only inhabit those areas.

areas. As the figure clearly shows, geopolitically sensitive and overlapping areas also have high critical species intensity when compared with other global areas.

Figure 6: Critical species in geopolitically sensitive and overlapping areas (GOSA) and other areas (Other)

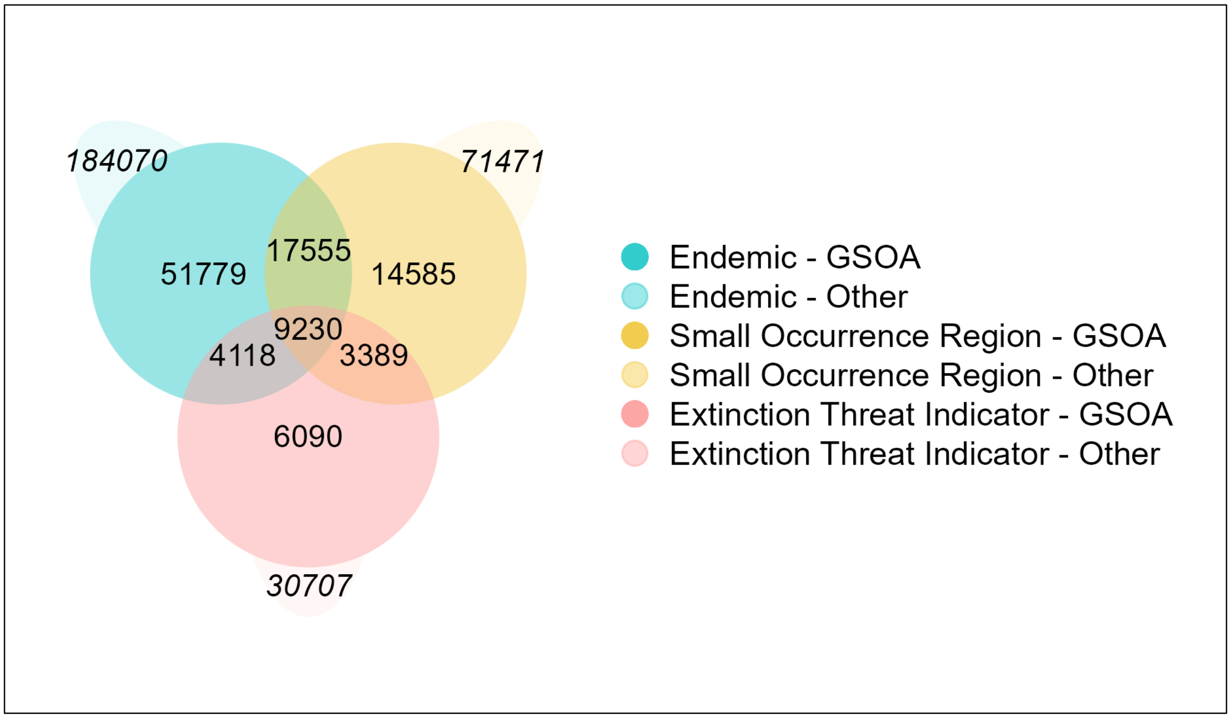
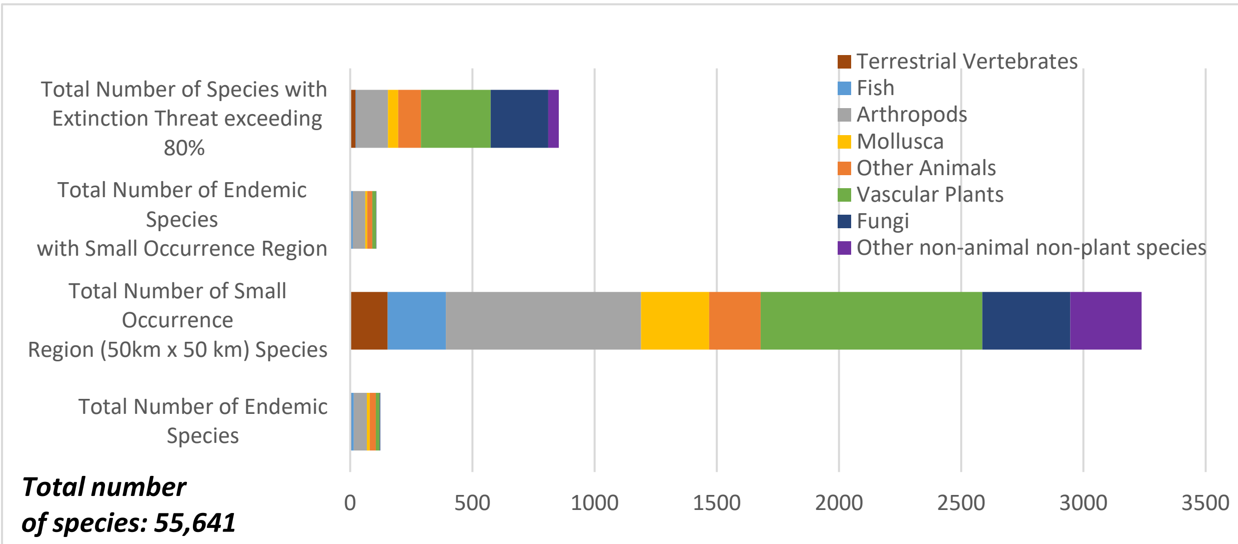


Figure 7 introduces another important dimension by showing that within each of the geopolitically sensitive and overlapping areas, representation of different taxa can vary widely. Particularly striking are variations in the representation of plants (green) and arthropods (gray) relative to representation for terrestrial vertebrates (brown) and fish (light blue). As we previously noted, our GBIF mapping exercise has greatly expanded geographic representation for plants and arthropods. In the current context, variations in this representation can provide important new information for the policy dialogue.

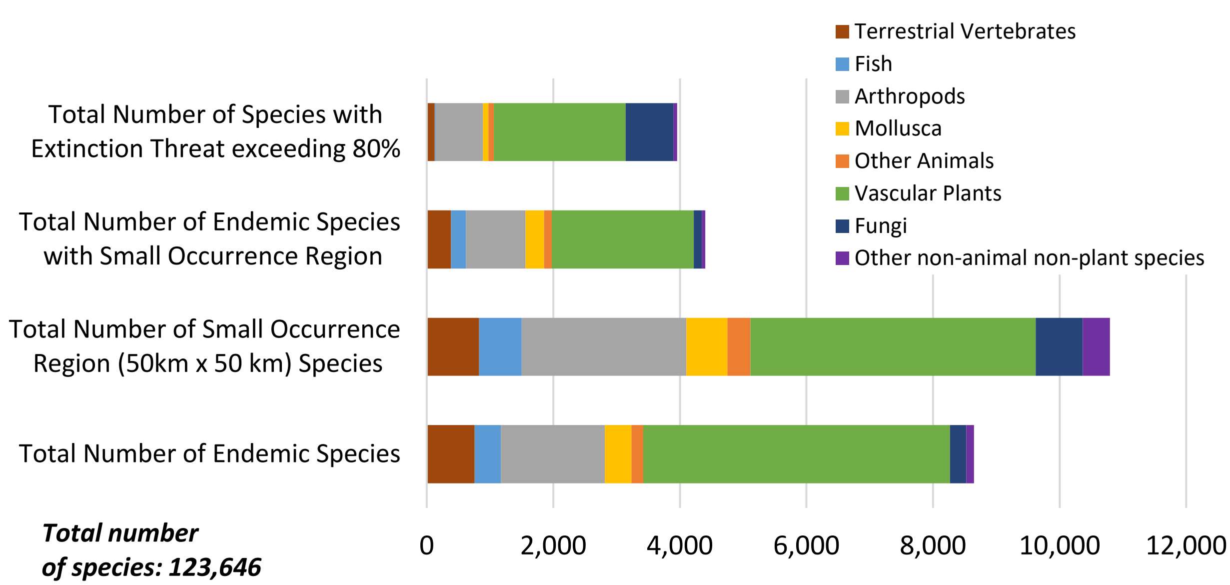
Figure 8 provides a global perspective by displaying the same charts for all species in the database. Comparison with Figure 7 shows a close resemblance to the distributions for Transboundary River Basins, with somewhat greater representation for molluscs and other animals in the full global distribution.

Figure 7: Taxon variations (species class/order) in GSOA

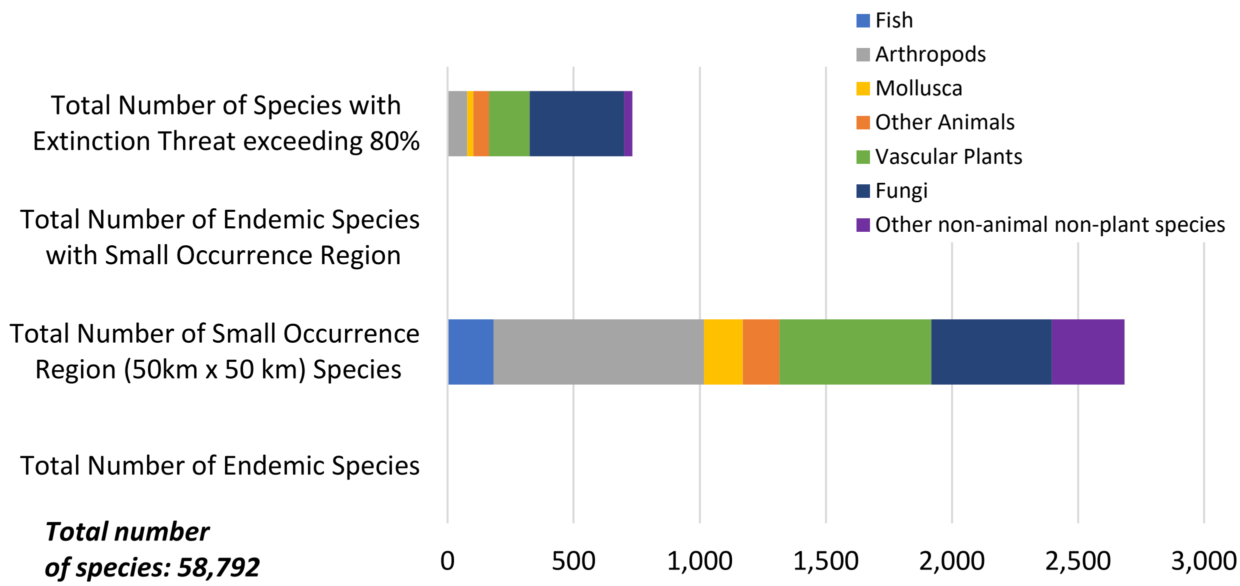
Non-determined Legal Status Areas



Fragile Conflict States



Marine Joint Regime



Transboundary River Basins

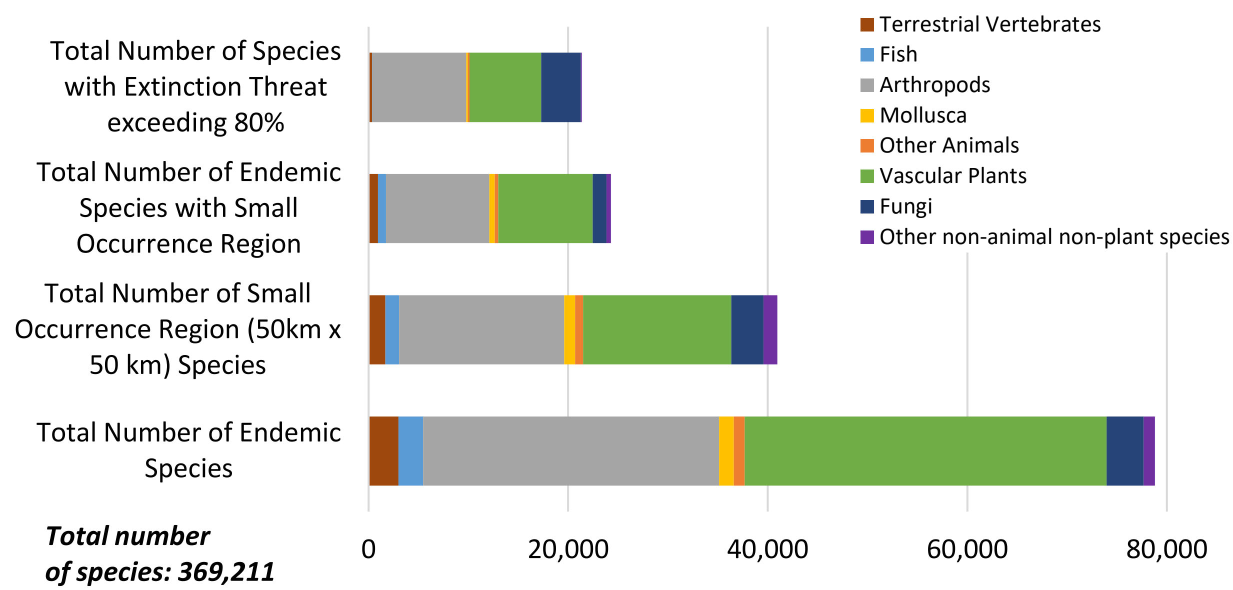


Figure 8: Global taxon variations

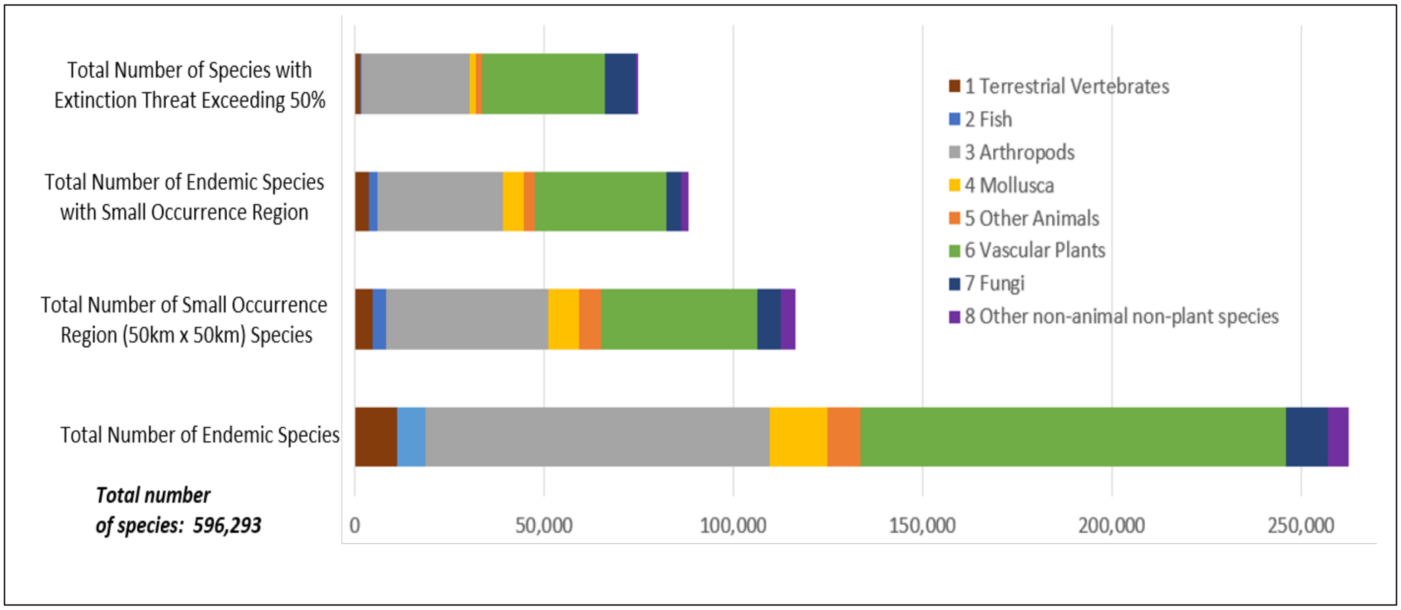


Table 3 summarizes the protection status of species in transboundary regions in analyses for international treaties and river basin organizations (RBOs). For species in critical categories (endemic, small range or high extinction threat indicator (ETI), between 4.9% and 9.6% are in basins that are not covered by international treaties, and between 12.8% and 29.9% are in basins not managed by RBOs. Between 4.9% and 9.4% are in basins not covered by treaties or RBOs.

This lack of governance coverage is especially troubling for endemic species with small occurrence regions (9.4% without treaties; 29.9% without RBOs; 9.3% with neither) and high ETIs (9.6% without treaties, 29.6% without RBOs; 9.4% with neither). The absence of governance structures in these regions means that conservation efforts are more difficult to coordinate, and species are at heightened risk from habitat destruction, overexploitation, and other environmental threats.

The findings in Table 3 highlight two major challenges for cooperation in transboundary river basins. First, even in areas where treaties or RBOs do exist, it is not guaranteed that species protection is a priority within these agreements. Many existing treaties and RBOs focus on issues such as water allocation and infrastructure development, but they may not explicitly address biodiversity conservation or the protection of species. This underscores the need for more targeted agreements within these frameworks that prioritize the protection of species, particularly those with small or restricted ranges that are highly vulnerable to environmental changes.

Second, in areas where no treaties or RBOs are in place, the absence of formal governance makes it more difficult to establish and enforce conservation measures. In these regions, there is a pressing need for alternative mechanisms to ensure that conservation stewardship responsibilities are both delegated and effectively monitored. This may involve the creation of ad hoc agreements between neighboring countries or the establishment of transboundary conservation initiatives that focus specifically on biodiversity protection. These initiatives could help fill the gaps where formal governance structures are lacking, promoting a more coordinated approach to conservation across national borders.

Ultimately, the results from Table 3 underscore the urgent need for international cooperation to ensure the protection of species in transboundary river basins. Without adequate governance frameworks, both at the treaty and RBO level, the survival of many species in these regions remains uncertain. Efforts to strengthen governance, prioritize species protection within existing agreements, and develop new mechanisms for regions without formal frameworks are critical for the long-term success of biodiversity conservation in these ecologically important areas.

Table 3: Transboundary river basins: species covered by treaties and river basin organizations

	Total Species	% Without Treaty	% Without RBO	% Without RBO or Treaty
Species Presence	369,211	4.93	12.78	4.87
Endemic	78,800	7.01	21.99	6.97
Small Occurrence Region	40,951	8.16	23.91	8.01
Extinction Threat Indicator (ETI) > 80	21,362	8.08	23.76	7.93
Endemic with Small Occurrence Region	24,288	9.38	29.85	9.28
Small Occurrence Region with ETI > 80	11,513	9.28	26.88	9.04
Endemic with ETI > 80	12,326	8.84	28.20	8.75
Endemic, Small Occurrence Region with ETI>80	8,339	9.57	29.57	9.44

Discussion

The data provided by the Global Biodiversity Information Facility (GBIF) can play a pivotal role in identifying and protecting vulnerable species, especially those located in fragile regions and ecosystems that span multiple political jurisdictions. One of the primary benefits of GBIF-type data is its ability to provide reliable and comparable information across borders, which is critical for the effective management of transboundary ecosystems and biodiversity.[9] For example, the Mekong River Basin, shared by six countries, relies on shared data to manage fisheries, water resources, and biodiversity effectively (Anh 2021; McPherson and Ropicki 2021). Similarly, the Central Albertine Rift, home to endangered species such as mountain gorillas, requires harmonized biodiversity data from Uganda, Rwanda, and the Democratic Republic of Congo to ensure conservation success (Plumptre, et al. 2010; Plumptre, et al. 2007). Variations in data collection methods, inconsistent monitoring standards, or the absence of cross-border data sharing can hinder ecosystem-level management efforts. The literature also emphasizes that standardized data collection improves the ability to track species trends, measure ecosystem health, and implement conservation strategies that address ecosystem-wide threats. Comparable data are also necessary for designing and implementing programs such as Transboundary Protected Areas. The estimates presented in this paper can help in setting up baselines for identifying priority areas for cooperative environmental protection.

In another potential domain for GBIF contributions, international and regional treaties play a critical role in conserving natural resources and biodiversity, particularly in transboundary river basins, international waters, and joint marine areas. These treaties establish a legal framework for cooperation among nations, fostering sustainable and equitable management of shared ecosystems while supporting biodiversity. For instance, the UN Convention on the Law of the Sea (UNCLOS) governs marine biodiversity in areas beyond national jurisdiction, promoting the conservation and sustainable use of marine resources (Klein 2005). Similarly, the Ramsar Convention facilitates the protection of wetlands, including transboundary water systems essential for biodiversity (Stroud et al. 2022). Regional treaties have also proven effective in managing specific environmental challenges. The Convention on the Protection of the Mediterranean Sea Against Pollution, known as the Barcelona Convention, exemplifies collaborative frameworks that address pollution and habitat degradation in marine environments (Raftopoulos 1992; Montefalcone et al, 2021).

In a related context, the Transboundary Freshwater Dispute Database[10] provides information on treaties across the world for more than 300 international river basins (McCracken and Wolf 2019). Turgul et al. (2024) provide an overview of major findings about transboundary water conflict and cooperation the last three decades. Since the 1960s, the environment has been increasingly highlighted in treaties (Giordano et al. 2014). However, Turgul et al. (2024), in their overview of major findings for transboundary water conflict and cooperation during the last three decades, note that more attention is needed to environmental issues. Several international river basin treaties incorporate biodiversity measurements or frameworks to monitor and

[9] Roberson et al. (2021) point out that multinational coordination is required, especially for marine species, given that nearly all species in their sample span multiple national jurisdictions.

[10] Product of the Transboundary Freshwater Diplomacy Database, College of Earth, Ocean, and Atmospheric Sciences, Oregon State University. Additional information about the TFDD can be found at: http://transboundarywaters.science.oregonstate.edu.

protect biodiversity. These measurements typically involve tracking the health of ecosystems, monitoring species populations, and assessing the impacts of water management activities.

River basin organizations (RBOs) and treaties provide additional mechanisms for promoting cooperative management of shared waters while addressing threats like pollution, over-extraction, and habitat destruction. RBOs can play important roles in maintaining ecological balance, preserving biodiversity, and enhancing the resilience of critical ecosystems in the face of climate change and human-induced pressures.

In the same vein, international river basin treaties can incorporate biodiversity considerations through monitoring species populations, ecosystem health, and habitat protection. Several important examples can be cited. (1) The Mekong Agreement integrates biodiversity monitoring into its Environment Programme, tracking species diversity, migration of fish (e.g., the Mekong giant catfish - *Pangasianodon gigas*), aquatic ecosystem health, and wetland conditions. The Agreement emphasizes the protection of wetlands and fisheries, along with sustainable species management. These goals reflect the broader vision of the Mekong River Commission (MRC), to ensure "an economically prosperous, socially just, environmentally sound, and climate resilient Mekong River Basin" (MRC, 2024). (2) The Nile Basin Initiative (NBI) includes biodiversity indicators to assess water management impacts on wetlands, ecosystems, and species. The NBI monitors pollution and focuses on the conservation of wetlands and aquatic ecosystems in water management plans. Critical species characteristics of the kind presented in Table 3 can play important roles in this context, including the identification of flagship species for Nile Basin wetland groups (NBI 2021). Article 6 of the Nile River Basin Cooperative Framework focuses on the protection and conservation of the Basin, including protecting and conserving biological diversity (NBI 2010). The NBI 10-Year Strategy identifies environmental sustainability as one of its six strategic priorities (Goal 4). (3) The Amazon Cooperation Treaty Organization (ACTO) promotes biodiversity conservation through monitoring forest, river, and wildlife status. Indicators include deforestation rates, species diversity, and ecosystem conditions. The Amazon Environmental Information Network (RAISG) tracks biodiversity and land-use trends, supporting sustainable resource management. (4) The Zambezi River Basin's Shared Vision (2004) integrates biodiversity monitoring into the Zambezi River Action Plan. The Zambezi River Authority (ZRA) monitors wetland health, riparian vegetation and fish populations, with a focus on migratory and endemic species (e.g. Kafue Lechwe, an endemic species of antelope in ZAMCOM 2008). The treaty advocates for integrated management to conserve ecosystems, limit pollution, and protect endangered species.

While these river basin initiatives have made indispensable contributions (Petersen-Perlman et al. 2017), further improvements are possible. For example, Liu et al. (2020) find that transboundary conservation efforts have predominantly concentrated on large mammals, often overlooking other species. Better information would also help, since a key element in the successful implementation of the treaties is the availability of reliable and comparable data for biodiversity monitoring, habitat assessments, and cross-border implementation. Reliable data underpin early warning systems, foster evidence-based decision-making, and inform the coordination of conservation efforts across borders. Moreover, such data are vital in addressing the impacts of climate change and human activities, where unilateral actions may have unintended transboundary effects.

In this context, we believe that GBIF data can play a critical role in expanding the information base for species that are critical for biodiversity conservation. As noted above, a major result of our species-mapping project has been a greatly expanded view of occurrence regions for invertebrates and plants that are endemic, inhabit small regions, or confront elevated extinction threats. The effectiveness of biodiversity protection can only be enhanced by introducing this information into the design and implementation of transboundary treaties, river basin initiatives, and stabilization measures for fragile/conflict states.

Biodiversity and natural resources also offer a unique, neutral platform for dialogue and confidence-building between divided groups, as evidenced by several cases worldwide. The Nile Basin Initiative encourages dialogue among Nile River countries, promoting integrated water resource management and sustainable development. The "Peace Parks" in Southern Africa, transboundary conservation areas like the Great Limpopo Transfrontier Park, have been created to promote peace and cooperation among Mozambique, South Africa, and Zimbabwe (Wolmer 2003). By focusing on shared environmental objectives, such initiatives help to build trust, reduce historical tensions, and foster a sense of shared responsibility among nations that may have previously experienced conflict. Similarly, in the Middle East, environmental issues have provided common ground for dialogue among Israelis, Palestinians, and Jordanians. Through initiatives like the Red Sea-Dead Sea Water Conveyance Project and joint efforts to rehabilitate the Jordan River, environmental peacebuilding has emerged as a valuable tool for fostering cooperation and understanding (Aggestam and Sundell 2016; Markel, Alster and Beyth 2013).

Recently, a report by the World Bank (2022)[11] has highlighted biodiversity as a valuable entry point for intergroup dialogue and cooperation. Again, however, reliable and comparable biodiversity data are critical for the success of such initiatives. Accurate data can provide common ground for all stakeholders by promoting a clear understanding of biodiversity status, trends, and threats. Such information helps depoliticize discussions by focusing on objective information that can serve collaborative activities. Consistent and comparable data can also support effective monitoring and evaluation of biodiversity initiatives, ensuring that progress can be measured and adjusted as needed.

Conclusion

Given the unprecedented loss of biodiversity, it is essential for every country and territory to engage in conservation efforts to minimize further degradation and its impact on human well-being. Consistent, dependable, location-specific biodiversity data are crucial for identifying species at risk, monitoring environmental changes and informing policy decisions. These factors

[11] https://www.worldbank.org/en/topic/environment/publication/defueling-conflict-environment-and-natural-resource-management-as-a-pathway-to-peace

are particularly important in critical biodiversity areas that span national borders and fragile/conflict regions.

The estimates derived from GBIF data presented in this paper can serve as a valuable baseline for initiatives to address such problems. As thousands of new reports are added daily to the GBIF repository, a steadily-expanding inventory of up-to-date species occurrence maps can provide a critical, objective information resource for tracking biodiversity trends, evaluating conservation progress, and developing effective policies.

This critical information can also provide more general benefits, because initiatives that focus on biodiversity can serve as valuable entry points for general dialogue and confidence-building among divided groups, fostering collaboration towards shared goals with multiple benefits (such as peace dividends). Threats to natural resources, including biodiversity loss, may be less politically sensitive and overlapping than other issues, and can encourage stakeholders to consider long-term perspectives on issues that transcend political boundaries.

Finally, we should re-emphasize the importance of new species knowledge in this context. This exercise has shown that the GBIF data can support a large expansion of information about species (particularly invertebrates and plants) that have critical importance for biodiversity conservation because they are endemic, inhabit small areas, or face serious extinction threats. While past transboundary and fragile/conflict state initiatives have made important contributions to environmental protection, they can be enhanced by moving beyond the traditional focus on vertebrate species to a much broader set of species that warrant protection.

Data Availability

The data can be downloaded from
https://datacatalog.worldbank.org/search/dataset/0066034/global_biodiversity_data

For a related blog, see Bridging Conflicts and Protecting Global Biodiversity

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